COMP-251 Practice Exam

Duration: 90 Minutes **Total Marks: 40**

Section A: Graph Algorithms

Question 1 (10 marks) - graphs

Given a directed weighted graph G = (V, E) with non-negative edge weights and a set of k designated hub vertices, design an algorithm to compute the shortest path from source s to target t that passes through at least one hub vertex. State its worst-case time and space complexity.

Question 2 (10 marks) - dynamic_programming

A data stream arrives in n intervals with processing costs c[1..n] and cooldown step k. Formulate a dynamic programming recurrence to maximize net throughput without scheduling two intervals within k steps of each other.

Section A: Trees and Hashing

Question 1 (10 marks) - trees

Write a Python function `is_height_balanced(root)` that checks in O(n) time whether a binary tree is height-balanced, returning both the balance status and subtree height in a single post-order traversal.

Section B: Dynamic Programming

Question 1 (10 marks) - hashing

Compare open addressing with linear probing against separate chaining when the load factor alpha approaches 0.9. State the expected probe count and cache locality trade-offs.